

The Prompt Ladder: Iterative Prompt Engineering (Week 2)

Track: General AI Fluency
Type: Assignment Deliverable

Phase: Foundations
Workload: 2 Hours

BASELINE: WEAK INITIAL PROMPT

Prompt: "Help me with marketing."

OUTPUT EXCERPT:
"Marketing is essential for any business! Here are 10 key strategies: 1. Social Media Marketing (post regularly on LinkedIn, X, and Instagram). 2. Content Marketing (start a blog and write informative articles). 3. SEO (optimize your website for relevant keywords). 4. Email Marketing (send weekly newsletters)..."

VERSION 1: ADDED LAYER — CLEARER GOAL

Prompt: "Write a promotional launch email for a new developer tool that automatically generates API documentation."

OUTPUT EXCERPT:
"Subject: Introducing AutoDocs - The Ultimate API Documentation Tool!
Hi [Name], Are you tired of writing API docs? Say hello to AutoDocs! Our revolutionary tool automatically scans your code and builds beautiful, interactive API documentation in seconds. Whether you're a startup or an enterprise, AutoDocs saves you hours of boring work. Sign up today for a 14-day free trial!"

What Changed in Prompt	What Improved in Output	What Still Failed	What I Would Try Next
I picked a real task: writing a launch email for an API documentation tool.	It stopped giving generic textbook advice and gave an actual email draft.	It sounds like pushy sales talk ('Say hello!', 'Revolutionary!') that real engineers ignore.	Tell it who we are writing to so the tone feels grounded.

VERSION 2: ADDED LAYER — DEFINED AUDIENCE

Prompt: "Write a promotional launch email for a new developer tool that automatically generates API documentation. Target audience: Senior Backend Engineers and Tech Leads."

OUTPUT EXCERPT:
"Subject: Stop writing API docs manually - let AutoDocs handle it
Hi [Name], Writing and updating API documentation takes time away from shipping features. AutoDocs integrates into your CI/CD pipeline, inspects your code annotations, and generates OpenAPI-compliant docs on every commit. It keeps your API specs up to date without interrupting your workflow. Try it today."

What Changed in Prompt	What Improved in Output	What Still Failed	What I Would Try Next
I added target audience: Senior Backend Engineers and Tech Leads.	It dropped the loud sales pitch and focused on engineering topics like CI/CD and OpenAPI specs.	It feels vague because it doesn't mention actual tool details like language support or architecture.	Provide concrete technical details about how the tool works.

VERSION 3: ADDED LAYER — REAL CONTEXT

Prompt: "Write a promotional launch email for a new developer tool that automatically generates API documentation. Target audience: Senior Backend Engineers and Tech Leads. Context: The tool is called 'DocForge'. It works via a lightweight CLI, supports Python (FastAPI/Django) and Node.js (Express), is completely open-source, and runs locally without sending code to cloud servers."

OUTPUT EXCERPT:

"Subject: DocForge: Open-source, local API doc generator for Python & Node
Hi team, Keeping API docs synced with codebase changes is a pain. We built DocForge - a lightweight CLI tool for Python and Node.js developers. It scans your codebase locally and builds interactive OpenAPI documentation during build time.

- 100% Local & Private: Your code never leaves your infrastructure.
- Zero Config CLI: Run `docforge init` to auto-generate docs.
- Open Source: MIT licensed and free to self-host.

Check out our GitHub repository to install the CLI or read the quickstart guide."

What Changed in Prompt	What Improved in Output	What Still Failed	What I Would Try Next
I added real details: tool name (DocForge), CLI setup, languages, and local privacy rules.	The email feels specific. Mentioning local execution directly addresses security concerns.	The layout looks like a basic bulleted sales list instead of a direct developer note.	Set strict rules on word count, bullet removal, and tone.

VERSION 4: ADDED LAYER — CONSTRAINTS

Prompt: "Write a promotional launch email for a new developer tool that automatically generates API documentation. Target audience: Senior Backend Engineers and Tech Leads. Context: The tool is called 'DocForge'... Constraints: Max 120 words. No bullet points. No hype adjectives (e.g., 'revolutionary'). Must sound like a peer engineer writing a quick update."

OUTPUT EXCERPT:

"Subject: DocForge - Open-source API doc CLI (Python & Node)
Hey, Quick update: we built a small CLI tool called DocForge to solve our own headache with stale API docs. It parses FastAPI, Django, and Express routes directly from your source code and generates OpenAPI docs locally. Because it runs on your machine or CI pipeline, your codebase never leaves your local environment or hits external servers. It's open-source (MIT) and takes about two minutes to test. If you want to try it out: `npm i -g docforge-cli` or check the source on GitHub: [Link]. Feedback or PRs are welcome."

What Changed in Prompt	What Improved in Output	What Still Failed	What I Would Try Next
I added strict limits: max 120 words, no bullets, no hype words, and a peer-to-peer tone.	It reads fast, sounds totally natural, and feels like a practical update from a teammate.	It does not clearly distinguish between an internal note and an external launch announcement.	Test providing a concrete tone example to guide output.

VERSION 5: ADDED LAYER — EXAMPLES OF WHAT GOOD LOOKS LIKE (HONEST REGRESSION)

Prompt: "...Example of tone to emulate: 'Hey team, just shipped v1.2 of our internal build tool. Added support for parallel builds and fixed memory leak. Pull latest main branch.'"

OUTPUT EXCERPT:

"Subject: DocForge CLI v1.0 shipped
Hey team, Just shipped v1.0 of DocForge, our local API doc CLI. Added support for FastAPI, Django, and Express routes. It parses code locally on your machine so no source code hits external cloud servers. Fixed the issue where nested route params wouldn't resolve properly. It's MIT licensed. Install with `npm i -g docforge-cli` or pull the latest main branch from GitHub. Let me know if you run into bugs."

What Changed in Prompt	What Improved in Output	What Still Failed	What I Would Try Next
I added a short example showing a super casual engineering update style.	Nothing improved. This change actually made the output worse.	The example was too internal. It turned a product launch into an internal bugfix note, losing new users.	Drop the internal example and enforce explicit quality criteria instead.

FINAL CLEAN, REUSABLE PROMPT

Engineered prompt structured with full context, explicit constraints, and verification criteria for independent execution:

[TASK]

Write a product launch announcement email for a developer tool.

[TARGET AUDIENCE]

Senior Backend Engineers, DevOps Engineers, and Tech Leads.

[PRODUCT CONTEXT]

- Product Name: DocForge
- Core Function: Lightweight CLI tool that automatically generates OpenAPI-compliant API documentation from codebase annotations.
- Tech Stack Supported: Python (FastAPI/Django) and Node.js (Express).
- Key Differentiators: 100% Open-Source (MIT), runs completely locally during CI/CD or build step, zero data sent to external servers.

[STYLE & TONE CONSTRAINTS]

- Length: Maximum 120 words.
- Format: Plain text paragraphs only. Strictly NO bullet points, bold headers, or numbered lists.
- Tone: Peer-to-peer engineering update. Honest, direct, and pragmatic.
- Banned Words: "Revolutionary", "game-changing", "synergy", "seamless", "cutting-edge", "next-gen", "supercharge".

[QUALITY & VERIFICATION CRITERIA]

1. Clearly state the exact problem solved in the first two sentences.
2. Address data privacy and security explicitly (local execution, no cloud leakage).
3. End with a single, clear action step (CLI install command and GitHub repository link).